

A REPORT

on

Industrial Training – I

INTERNSHIP REPORT



Submitted in Partial Fulfillment of the requirements
For the Degree of Bachelor of Technology in CSE

By

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Netaji Subhas Institute of Technology (NSIT), Patna
Bihar Engineering University (BEU), Patna

Academic Session: 2024 – 2028

Training Completed at: C-DAC, Patna
Training Period: 22nd December 2025 – 3rd January 2026

CERTIFICATE OF COMPLETION

This is to certify that the report entitled "Industrial Training — I" is a bonafide record of the work carried out by **Saniya Suman** (Registration No.: 24105103036), a student of Bachelor of Technology in Computer Science and Engineering (CSE) during the academic session 2024–2028 in the Second Year, Third Semester at Netaji Subhas Institute of Technology (NSIT), Patna.

This report has been submitted in partial fulfillment of the requirements for the award of the Bachelor of Technology (B.Tech) degree under Bihar Engineering University (BEU), Patna.

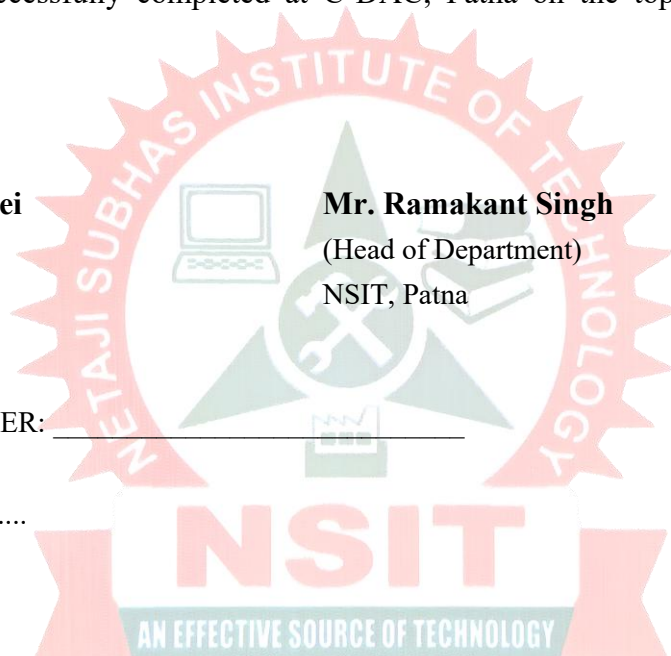
The training was successfully completed at C-DAC, Patna on the topic "**Core Java (OOP Concepts)**".

Dr. Jyotirmayee Dalei
(Academic Dean)
NSIT, Patna

Mr. Ramakant Singh
(Head of Department)
NSIT, Patna

EXTERNAL EXAMINER: _____

Date:



Acknowledgement

I take this opportunity to express my heartfelt gratitude to all those who have supported and guided me throughout my Industrial Training — I, enabling me to successfully complete this report entitled "Industrial Training — I".

Firstly, I would like to extend my sincere thanks to Netaji Subhas Institute of Technology (NSIT), Patna, for providing me with the opportunity to undertake this training, which has been a valuable learning experience in my academic journey.

I would also like to express my sincere gratitude to C-DAC, Patna, for organizing this training program on "Library Management System using Core Java" and for providing excellent learning resources and practical exposure.

I am highly thankful to the respected faculty members and trainers, especially Mr. Vaibhav Patle and Ms. Nishu, for their continuous guidance, support, and valuable insights throughout the training period.

I would also like to thank Dr. Jyotirmayee Dalei (Academic Dean) and Mr. Ramakant Singh (Head of Department), NSIT Patna, for their encouragement and support.

Finally, I extend my deep appreciation to my family, friends, and well-wishers, whose constant motivation and support have been a source of strength in successfully completing this training and report.

Place: Patna

Date:

Student Name: **Saniya Suman**

Registration No.: 24105103036

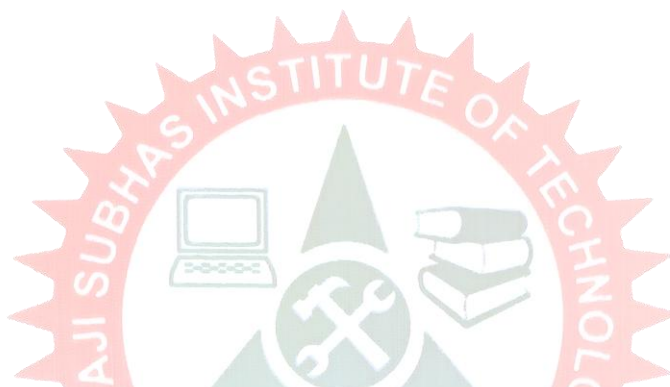
List of Abbreviations

OOP	—	Object-Oriented Programming
JDK	—	Java Development Kit
JVM	—	Java Virtual Machine
JRE	—	Java Runtime Environment
LMS	—	Library Management System
SDLC	—	Software Development Life Cycle
GUI	—	Graphical User Interface
CLI	—	Command Line Interface
API	—	Application Programming Interface
IDE	—	Integrated Development Environment
I/O	—	Input/Output
CRUD	—	Create, Read, Update, Delete
CPU	—	Central Processing Unit
RAM	—	Random Access Memory
PRN	—	Participant Registration Number



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CHAPTER 1: INTRODUCTION

1.1 Purpose of the Training

The purpose of this Industrial Training — I was to bridge the gap between theoretical knowledge and practical implementation. In college, we learn many concepts, but this training helped me understand how these concepts are actually used in real-life situations.

This training aimed to provide hands-on experience in programming using Core Java. It helped me improve my coding skills and understand how to write clean and efficient programs. I also learned how different concepts of Object-Oriented Programming (OOP) like classes, objects, inheritance, abstraction, and polymorphism are applied in real software development.

Another important purpose of this training was to develop problem-solving ability and logical thinking. By working on practical tasks and building a real Library Management System project, I gained confidence in coding and debugging errors.

Overall, this training helped me understand the basics of software development, improved my technical knowledge, and prepared me for future academic projects and industry work.

1.2 Objectives

The main objectives of the training were:

- To understand the fundamentals of Java Programming Language
- To learn and implement Object-Oriented Programming (OOP) concepts
- To gain practical knowledge of Core Java features such as classes, objects, inheritance, abstraction, and polymorphism
- To develop a real-world Library Management System project
- To develop problem-solving and coding skills
- To gain exposure to real-time project development
- To enhance technical and logical thinking abilities

1.3 Overview of Training Program

The Industrial Training — I was conducted at C-DAC, Patna for a duration of two weeks, from 22nd December 2025 to 3rd January 2026. The training sessions were held daily from 9:00 AM to 1:30 PM.

The training focused on "OOP's Concepts using Core Java" and included both theory and practical sessions. In the beginning, basic concepts of Java were taught, such as data types, operators, and control statements. After that, we learned important Object-Oriented Programming (OOP) concepts like classes, objects, inheritance, polymorphism, and encapsulation.

As the training progressed, more advanced topics were covered, including interfaces, exception handling, file handling, collections framework, and multithreading. Each topic was explained

with examples and practical coding, which made it easier to understand and apply in our project — the Library Management System.

The sessions were conducted by experienced faculty members, **Mr. Vaibhav Patle** and **Ms. Nishu**, who guided us throughout the training. At the end of the training, we worked on the Library Management System project and gave a final assessment to test our understanding.

1.4 Company Profile: C-DAC, Patna

The Centre for Development of Advanced Computing (C-DAC) is a well-known research and development organization in India. It works under the Ministry of Electronics and Information Technology (MeitY), Government of India.

C-DAC was established to develop advanced technologies in the field of computing. Over the years, it has played an important role in the growth of Information Technology in India. It is especially known for its work in supercomputing, software development, and technical education.

Location

The training was conducted at C-DAC, Patna, located in Bihar, India. This center provides a good learning environment with proper labs, systems, and experienced faculty members.

Services and Areas of Work

C-DAC works in many advanced areas of technology:

- **High-Performance Computing (HPC): Development** of supercomputers for research and scientific work
- **Software Development:** Designing and developing software solutions for various industries
- **Cyber Security:** Working on data protection, security systems, and ethical hacking
- **Artificial Intelligence (AI) & Machine Learning (ML):** Developing smart systems and automation technologies
- **Embedded Systems & IoT:** Working on hardware-based programming and smart devices
- **Education and Training:** Providing various training programs for students and professionals

Organizational Structure

- **Director General:** Overall head of C-DAC
- **Centre Director:** Head of a specific center (e.g., Patna)
- **Project Managers / Program Coordinators:** Manage projects and training programs
- **Faculty / Trainers:** Teach students and provide guidance
- **Technical Staff:** Help in lab work and system management During the training, the sessions were conducted by Mr. Vaibhav Patle and Ms. Nishu, who guided us in both theory and practical work

CHAPTER 2: TRAINING DETAILS

2.1 Duration of Training

The Industrial Training — I was conducted for a period of two weeks, from 22nd December 2025 to 3rd January 2026 at C-DAC, Patna.

The training sessions were held daily from 09:00 AM to 01:30 PM.

This duration was very helpful to understand both basic and advanced concepts of Java programming in a structured way.

2.2 Department

The training was conducted under the Software Development / Training Department of C-DAC, Patna. This department focuses on providing technical knowledge and practical skills to students in the field of programming and software development. It also helps students to understand how real-world software systems are developed.

2.3 Tools and Technologies Used

During the training, I worked with different tools and technologies related to Core Java:

- Core Java (JDK 8+): Used for writing and running Java programs
- Object-Oriented Programming (OOP): Concepts like abstract classes, interfaces, inheritance, polymorphism, and encapsulation
- Java Collections Framework: ArrayList used to manage book and member data
- Exception Handling: Custom exceptions (LibraryException, BookNotFoundException) to handle runtime errors
- File Handling: FileWriter, BufferedReader, PrintWriter for persistent data storage
- IDE (Eclipse / IntelliJ IDEA): Used to write and execute Java programs

2.4 Training Methodology

The training program was designed in a simple and effective way. Each topic was first explained in theory, and then practical examples were given:

- Trainers explained concepts using real-life examples
- After explanation, coding was done step-by-step
- Students were encouraged to practice on their own
- Doubts were cleared regularly
- A final project (Library Management System) was built by applying all concepts

2.5 Daily / Weekly Activities

Week 1 (22nd December – 27th December 2025)

In the first week, the focus was on basic concepts of Java:

- Introduction to Java Programming Language
- Data Types and Operators
- Flow Control Statements (if-else, loops)
- OOP Concepts (Class and Object)

- Constructor and Parameter Passing
- Inheritance in Java
- Abstract Classes and Interfaces
- Packages and Import Statements
- Introduction to Arrays

This week helped me build a strong foundation in Java.

Week 2 (29th December 2025 – 3rd January 2026)

In the second week, more advanced topics were covered:

- Wrapper Classes and String Handling
- Exception Handling (Custom Exceptions)
- File Handling (FileWriter, BufferedReader, PrintWriter)
- Collections Framework (ArrayList)
- Multithreading and Inner Classes
- Project Work: Library Management System
- Final Assessment

During this week, I got practical exposure and worked on the Library Management System project with my team.

2.6 Overall Experience

The training experience at C-DAC Patna was very good. The sessions were interactive, and the trainers were supportive. I was able to learn new concepts and improve my practical skills. This training helped me become more confident in Java programming and prepared me for future academic projects and real-world applications.



Fig. 1: Group Photo of Training Participants at C-DAC Patna

CHAPTER 3: PROJECT WORK

3.1 Project Title

Library Management System

(Console-Based Application using Core Java)

3.2 Introduction to the Project

During the training, we developed a Library Management System using Core Java. It is a console-based application, which means it runs on the command prompt and does not use any graphical interface. The system works in a menu-driven way where users can register, log in, and perform various library operations easily.

The main purpose of this project is to manage library operations such as book issuing, returns, member management, and fine calculation. Instead of using a database, all data is stored in simple text files (users.txt, book_issues.txt, library_records.txt). This helped us understand file handling in Java and how data can be saved and managed without using complex database systems.

This project helped us understand how real-world applications are developed using Java. We learned how to apply OOP concepts — including abstract classes, inheritance, interfaces, polymorphism, encapsulation — file handling, collections, and exception handling in one integrated project.

3.3 Objectives of the Project

- To apply Core Java and OOP concepts in a real library management project
- To implement abstract classes, inheritance, and polymorphism through Book class hierarchy
- To implement interface-based design using LibraryOperations interface
- To implement a secure login and registration system with file persistence
- To understand file handling for persistent data storage
- To improve coding, problem-solving, and teamwork skills

3.4 Team Structure

This project was developed by a team of 3 members, where each member had a specific role and responsibility. The work was divided into different modules so that the project could be completed in an organized and efficient way.

S.No.	Name	PRN	Role / Module
1	Saniya Suman	STT-25128071914	Main Application (LibraryProject.java) + Interface

			Design (LibraryOperations.java) + Library Business Logic (LibraryService.java)
2	Shivani Kumari	STT-25128071916	Book Model Classes (Book.java) — Abstract class+ Login System (LoginServices.java) + Fiction, NonFiction, Reference, Textbook
3	Tanisha Kumari	STT-25128071921	File Handling (FileHandler.java) + Custom Exceptions (LibraryException.java)

Working in a team helped us understand how real software projects are developed in parts. Each member focused on their task, and finally all modules were combined to build the complete system.

3.5 Project Structure

The project follows a proper modular structure where different functionalities are divided into separate Java files. This makes the system easy to understand, manage, and maintain.

Directory Structure:

```

PROJECT1/
├── LibraryProject.java      [Saniya Suman]      Main menu-driven application
├── Book.java               [Shivani Kumari]      Abstract Book + 4 book type
├── classes
│   ├── LibraryService.java [Tanisha Kumari]      Library business logic &
│   │   operations
│   ├── LoginServices.java  [Tanisha Kumari]      Member registration & login
│   ├── FileHandler.java    [Tanisha Kumari]      File I/O (billing & book
│   │   records)
│   ├── LibraryOperations.java [Saniya Suman]      Interface for library
│   │   operations
│   └── LibraryException.java [Tanisha Kumari]      Custom exceptions
├── Data Files (auto-generated):
│   ├── users.txt           Login credentials
│   ├── book_issues.txt     Active booking records
│   └── library_records.txt Transaction/billing history

```

Class & Method Interaction Flow:

1. User launches LibraryProject.main()
2. LoginServices.Register() / LoginServices.login()
3. On successful login → LibraryService.menu()
4. issueBook() → Book.issue(memberName, days)
5. FileHandler.saveBookingInfo(name, days, type)
6. Optional: Book.addFine(amount)
7. returnBook() → Book.returnBook(actualDays)
8. Book.calculateFine() = (fine × overdueDays)
9. FileHandler.saveFine(memberName, totalAmount)
10. FileHandler.deleteBookingInfo(roomNumber)

```
11. showAvailableBooks() → Book.availability()
12. LibraryException / BookNotFoundException on errors
```

3.6 Technologies Used

- Core Java (JDK 8+)
- Object-Oriented Programming: Abstract classes, interfaces, inheritance, polymorphism
- Collections Framework (ArrayList<Book>, ArrayList<User>)
- Exception Handling (Custom LibraryException, BookNotFoundException, InputMismatchException)
- File Handling (FileWriter, PrintWriter, BufferedReader, FileReader)
- IDE: Eclipse / IntelliJ IDEA

3.7 System Features

The system provides various features that help in managing library operations in an easy and efficient way:

Feature 1: User Authentication System

- Register new users with username and password validation (min. 6 characters)
- Secure login with a maximum of 3 attempts; account blocked after failed attempts
- User credentials stored persistently in users.txt
- Default admin account pre-loaded (admin / admin334)

Feature 2: Book Management – 4 Book Types

Book Type	Book IDs	Price / Night	Description
Fiction	B001 – B020	₹ 5 / day fine	20 books – novels, stories, drama
Non-Fiction	B021 – B040	₹ 10 / day fine	20 books – biographies, history, science
Reference	B041 – B055	₹ 20 / day fine	15 books – encyclopedias, dictionaries
Textbook	B056 – B075	₹ 15 / day fine	20 books – academic & curriculum books

Feature 3: Book Issuing (Check-Out)

- View all books with type, availability status, and details
- Issue any available book by entering book ID, member name, and number of days
- Booking information saved automatically to book_issues.txt
- Optional fine details can be added at the time of issue

Feature 4: Book Return & Fine Calculation

- Member can return book by entering book ID
- Actual days stayed can differ from booked days (early checkout supported)
- Refund calculated automatically for unused days
- Total Bill = (Price per Night × Days Stayed) + Service Charges
- Bill saved to library_records.txt with date/time stamp
- Booking entry removed from book_issues.txt on checkout

Feature 5: Fine Management

No.	Service	Charge
1	Overdue Fine	₹ 5/day
2	Lost Book Penalty	₹ 500
3	Damage Charge	₹ 200

Feature 6: File Handling System

All data is stored persistently in text files:

- users.txt → Login credentials for registered users
- book_issues.txt → Active issue records (book ID, member name, days, book type)
- library_records.txt → Transaction records (member name, fine amount, date/time)

3.8 Working of the Project

1. User launches the application — Welcome screen and main menu are displayed
2. User registers a new account or logs in with existing credentials (max 3 attempts)
3. On successful login, the Library Menu is displayed with 5 options
4. User can view all books or only available books
5. Member issues a book by providing member name and number of days; optional fine details are noted
6. Booking data is saved to book_issues.txt
7. On return, overdue days are calculated, fine is applied if late, and a detailed receipt is printed
8. Final transaction is saved to library_records.txt and booking entry is removed

CHAPTER 4: LEARNING OUTCOMES

4.1 Skills Gained

Through this project, several important technical and analytical skills were developed. Strong proficiency in Java programming was achieved, along with improved understanding of Object-Oriented Programming (OOP) concepts such as abstraction (abstract Book class), inheritance (Fiction, NonFiction, Reference, Textbook extending Book), polymorphism (overriding getBookName() and display()), and encapsulation (private fields with getters/setters).

Skills in problem-solving and logical thinking were enhanced by designing features like the add book, borrow book, return book. Additionally, experience was gained in debugging and error handling through custom exceptions (LibraryException and BookNotFoundException), which improved the ability to write efficient and error-free code.

4.2 Knowledge Acquired

This project helped in acquiring in-depth knowledge of how a complete software system is designed and implemented. A better understanding of Java Collections Framework (ArrayList) was gained for efficient room and user data management. Knowledge of file handling techniques was developed, enabling persistent storage of booking records, transaction logs, and user credentials.

The project also provided insights into interface-based design (LibraryOperations interface) and how it improves code modularity and scalability. Furthermore, it improved understanding of inner classes (User class inside LoginServices) and how to structure a multi-class Java project.

4.3 Practical Exposure

The project provided valuable hands-on experience in developing a real-world application. It offered practical exposure to the software development life cycle (SDLC), including planning, designing, coding, testing, and debugging. Experience was gained in working with real-time data processing, user input validation, and system interaction.

It also helped in understanding how different components of a system interact with each other — for example, how LibraryProject calls LoginServices for authentication and then delegates to LibraryService for all library operations, while FileHandler silently manages all persistence in the background. Overall, this project bridged the gap between theoretical concepts and practical implementation.

CHAPTER 5: CONCLUSION

5.1 Overall Experience

This project has been a highly enriching and insightful learning experience that contributed significantly to both technical growth and overall development. It provided an opportunity to practically implement theoretical concepts learned during the course, especially in Java programming and Object-Oriented Programming (OOP). By working on this Library Management System, a deeper understanding of concepts such as abstraction, encapsulation, inheritance, polymorphism, file handling, collections, and exception handling was achieved.

The entire process of developing the project — from requirement analysis, designing the class hierarchy, coding, testing, to debugging — helped in understanding how real-world applications are built. It improved the ability to break complex problems into smaller modules and solve them step by step. Working on features like add book, borrow book, return book, display book and user authentication gave practical exposure to real-time application scenarios.

During the project, several challenges were faced, such as handling errors during file operations, managing book correctly between borrowing and returning, and ensuring proper program flow with menu navigation. Overcoming these challenges helped in improving problem-solving skills, logical thinking, and debugging capabilities.

Overall, this experience has been extremely valuable in bridging the gap between academic knowledge and practical implementation, making it a strong foundation for future software development work.

5.2 Importance of Training

Industrial training is an essential part of an engineering curriculum as it provides real-world exposure and practical knowledge that cannot be fully gained through classroom learning alone. It plays a vital role in developing technical expertise and understanding how concepts are applied in real industry environments.

Through training, students gain hands-on experience in working with technologies, tools, and frameworks used in the industry. It helps in understanding the Software Development Life Cycle (SDLC), including planning, designing, coding, testing, and deployment. This practical exposure enhances the ability to work on real-time projects and prepares students for professional challenges.

Overall, industrial training is crucial for shaping a student's career. It provides a platform to apply theoretical knowledge, gain practical experience, and develop the skills required to become a competent software developer. This experience not only enhances technical abilities but also builds confidence and prepares for future career opportunities in the field of technology.

CHAPTER 6: REFERENCES

- Official Java Documentation:
<https://docs.oracle.com/javase/>
- GeeksforGeeks: <https://www.geeksforgeeks.org>
- W3Schools Java Tutorial:
<https://www.w3schools.com/java/>
- TutorialsPoint Java Guide:
<https://www.tutorialspoint.com/java/>
- Stack Overflow (for problem-solving and debugging support): <https://stackoverflow.com/>
- Class notes and lecture materials provided during academic sessions at C-DAC, Patna
- Practical lab manuals and assignments
- Online video lectures and tutorials on Java programming
- Previous knowledge and hands-on practice gained during diploma studies